

Pharmacology of Congestive Heart Failure (CHF)

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Session Objectives

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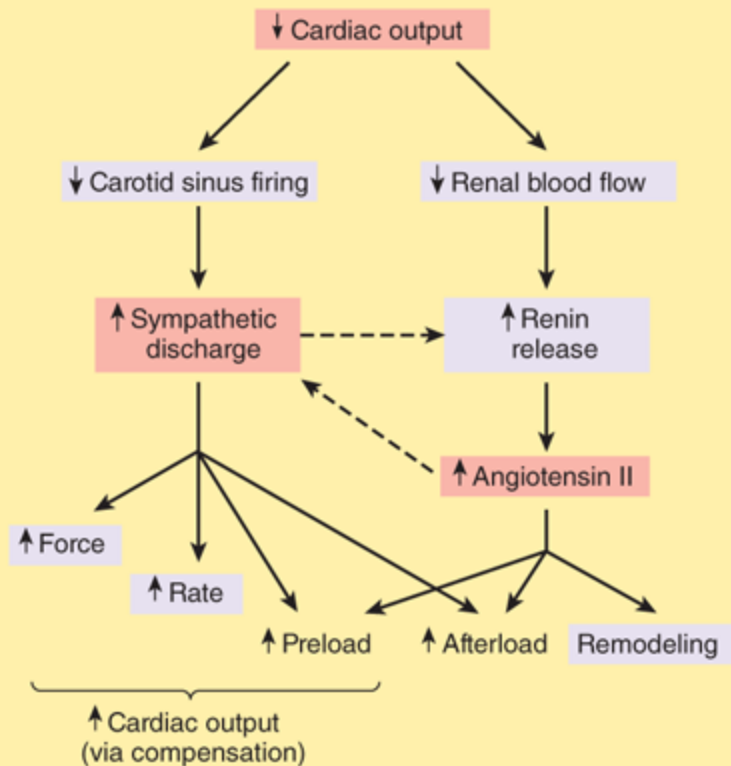
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- 1. Describe the specific goals of congestive heart failure (CHF) management.**
- 2. Correlate the pathophysiology of CHF with the drugs used for treatment.**
- 3. Explain how these drugs benefit the CHF patient.**
- 4. Examine the pharmacological mechanism of these drugs.**
- 5. Describe specific drug adverse effects.**

Pathophysiology of CHF

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Source: A.J. Trevor, B.G. Katzung, M. Kruidering-Hall: Katzung & Trevor's Pharmacology: Examination & Board Review, 11th Ed.
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- Inability of ventricle to pump blood into circulation
- Causes: recent MI, uncontrolled hypertension, arrhythmia.
- Drugs: Doxorubicin (intercalating agent, chemotherapeutic).
Thiazolidinediones “glitazones” (PPAR γ agonists for type II diabetes). Pioglitazone, Rosiglitazone
NSAIDs (vasoconstriction in kidney by prostaglandin blockade in the afferent arteriole)
- Results in cardiac remodeling

Cardiac Remodeling

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- **Post MI**
- **Stretch of muscle fibers**
- **Results in fibrosis, collagen, and myocyte hypertrophy**
- **Consequences: Reduced stroke volume and cardiac output, increased ventricle stiffness, decreased contractility**
- **Increase in sympathetics and renin-angiotensin-aldosterone axis (benefit of beta blockers, ACEI/ARBs/ARNI/spironolactone)**

CHF

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Left ventricular failure

- Heart can't maintain body's blood flow needs
- Pressure in pulmonary circulation increases
- Edema, dyspnea, orthopnea, fatigue, hypoxia, organ dysfunction

Right ventricular failure

- Right ventricle can't pump to lungs. Fluid build-up
- Ankle edema
- Hepatojugular congestion "nutmeg liver"



CHF

Dilated pupils, a sympathetic nervous system response

Skin pale, gray, or cyanotic

Dyspnea, SOB or early symptom from pulmonary congestion

Orthopnea, cannot breathe unless sitting up

Crackles, wheeze are adventitious breath sounds

Cough, frothy pink or white sputum

Decreased blood pressure stimulates sympathetic nervous system, which acts on heart to increase rate and increase force of contraction

Nausea and vomiting as peristalsis slows and bile and fluids back up into stomach

Ascites, fluid in peritoneal cavity

Dependent, pitting edema, in sacrum, legs

Anxiety, gasping from pulmonary congestion

Falling O₂ saturation

Confusion, unconsciousness from decreased O₂ to brain

Jugular vein distention from venous congestion

Infarct, may be cause of decreased cardiac output

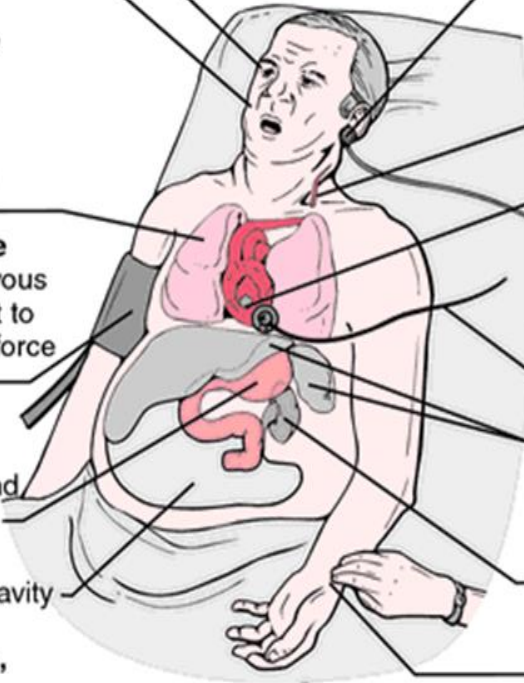
Fatigue, weakness from decreased cardiac output

S₃ gallop, tachycardia

Enlarged spleen and liver from venous congestion. This causes pressure on breathing

Decreased urine output

Weak pulse
Cool, moist skin



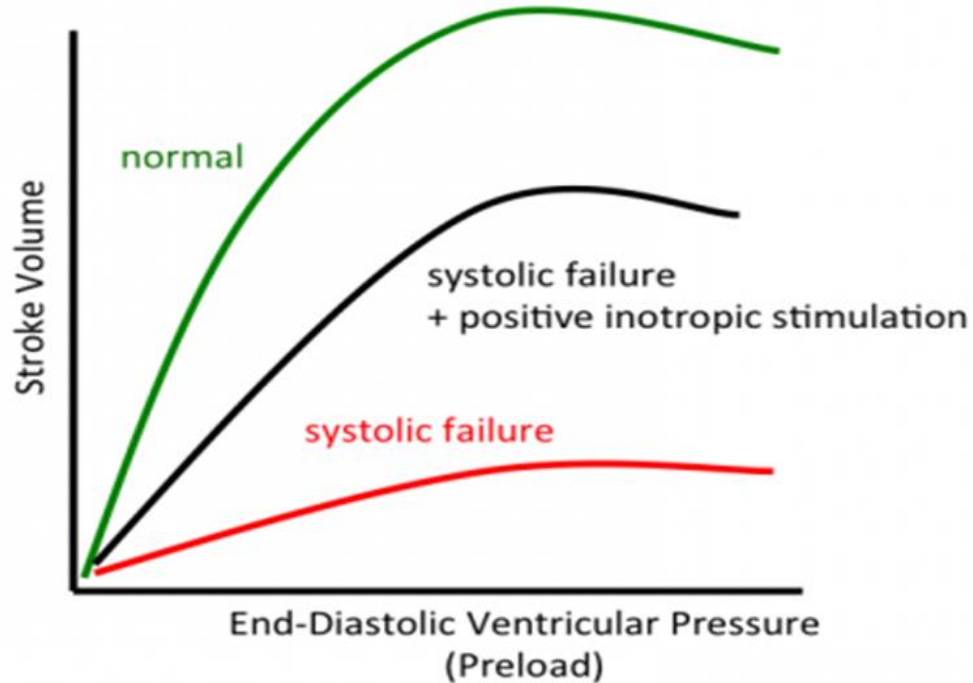
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Frank-Starling

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http://tmedweb.tulane.edu/pharmwiki/doku.php/treatment_of_heart_failure

CHF Management

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- **Diet, exercise, sodium restriction guidelines**
- **Decrease preload: diuretics, ACEIs/ARBS, venodilators**
- **Decrease afterload: ACEIs/ARBS, arteriodilators**
- **Improve contractility: digoxin, B₁ agonists, PDE3 blockers**
- **Decrease remodeling: ACEIs/ARBS, spironolactone, beta blockers.**
- **Other drugs: Antiplatelet or anticoagulants, antidiabetics, antihyperlipidemics, etc**

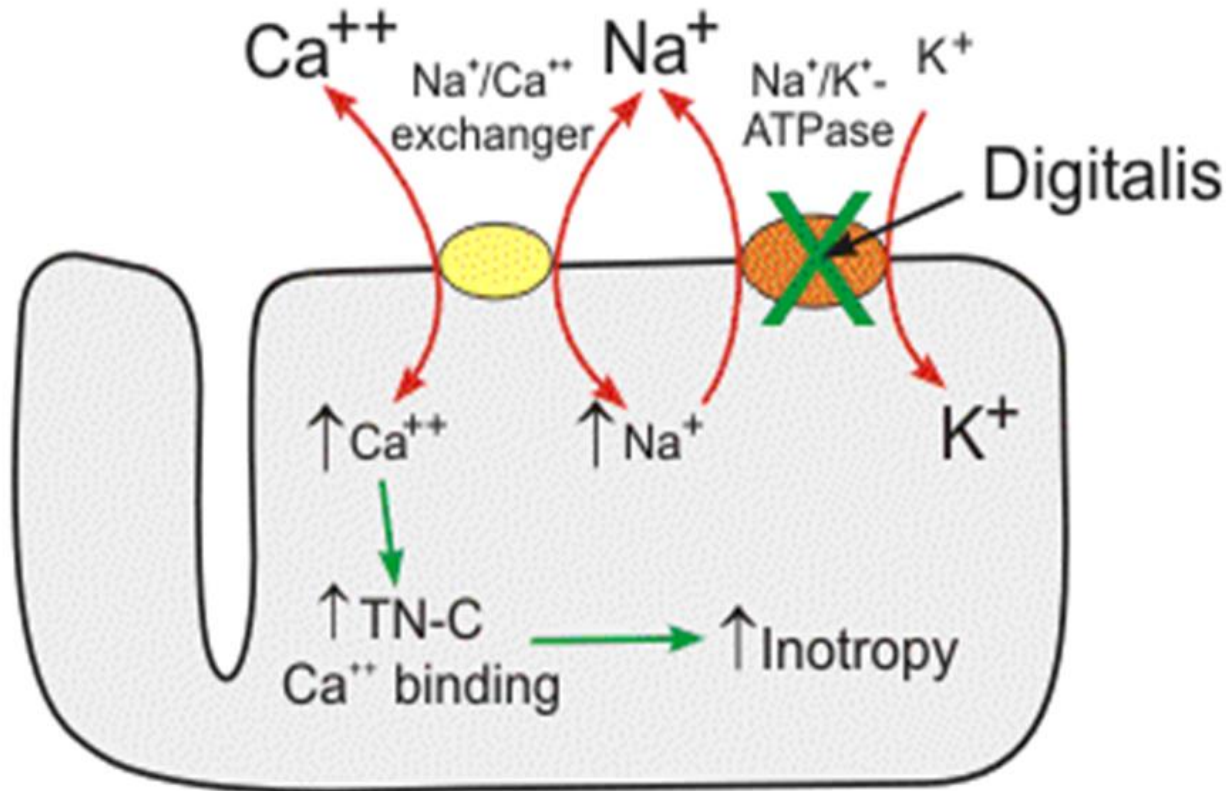
Positive Inotropic Drugs

- Increase contractility, mainly for acute decompensated heart failure
- Digoxin (digitalis, foxglove plant)
- Dobutamine (B_1 agonist, G_s mechanism).
- Dopamine (maintains GFR and renal blood flow)
- Milrinone (PDE-3 blocker)

Digoxin

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Digoxin

- Long $t_{1/2}$
- Negative chronotropic: increases parasympathetic tone (vagal)
- Increased afterdepolarizations (monitor K+)

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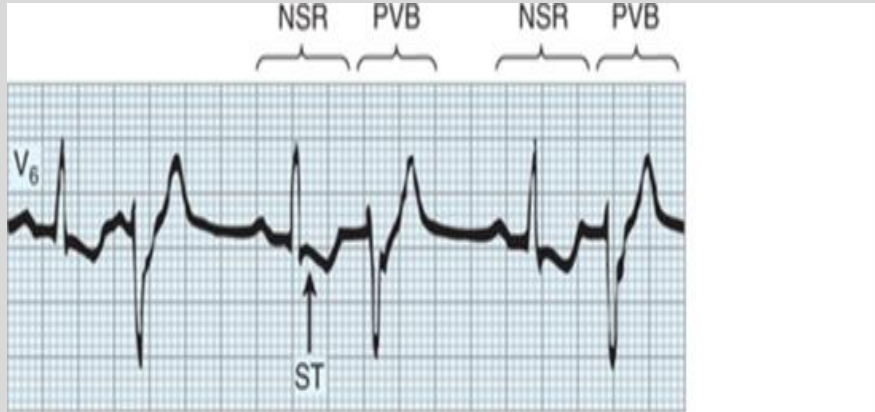


Digoxin

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- Increases PR (reduced AV node)
- ST depression (hockey stick configuration)



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Digoxin

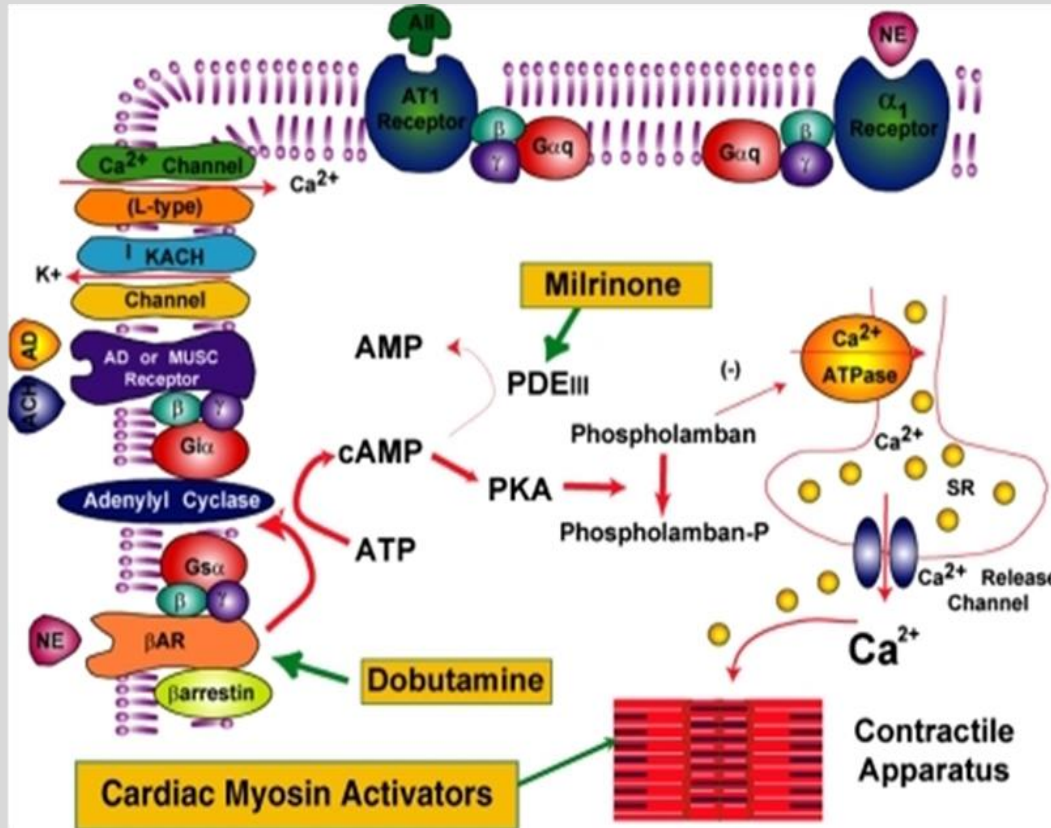
- N/V/D, anorexia
- Ventricular arrhythmias: High Ca, low K, Mg (monitor diuretic use)
- Yellow vision (halos)
- Gynecomastia (estrogenic)
- DI: antacids, cholestyramine (absorption) amiodarone (CYP inhibition), verapamil and quinidine (elimination)
- Renal impairment (reduced digoxin elimination)
- Antidote: DigFab, manage electrolytes. Give magnesium (torsades)



Dobutamine-Milrinone (increase cAMP)

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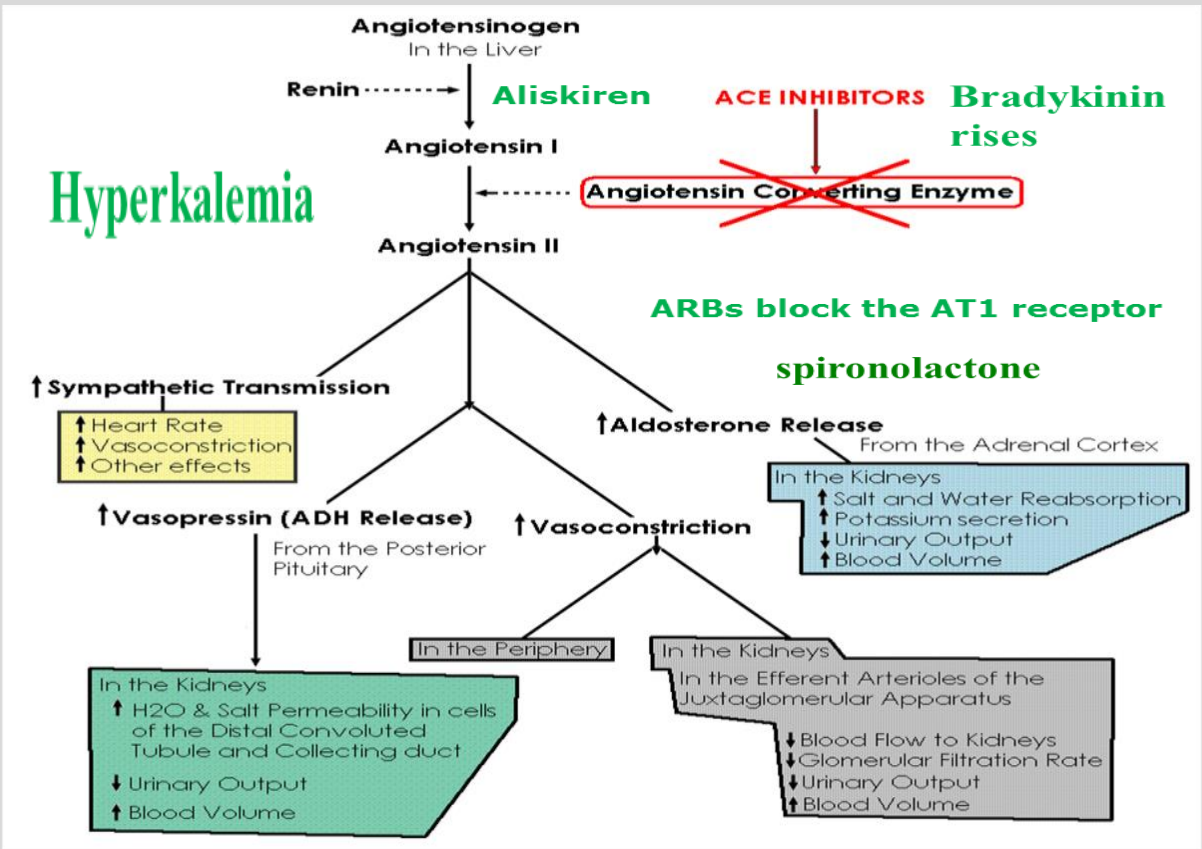
Vasodilators

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- Angiotensin-converting enzyme inhibitors (ACEI, -prils) and Angiotensin II receptor blockers (ARBs, -sartans). ****Chronic CHF*******
- Hydralazine
- Nitric oxide donors (nitroglycerin, isosorbide dinitrate, nitroprusside)

Renin-Angiotensin-Aldosterone System



RAAS Adverse Effects

- First-Dose Phenomenon
- Aldosterone absence: Hyperkalemia
- **Bradykinin: Cough, angioedema. Specific to ACEI. Replace with ARBs**
- Drug interactions (K⁺ sparing diuretics or supplements), NSAIDs.
- Renal artery stenosis (contraindication). But used to **reduce proteinuria in type II diabetics**
- Fetal renotoxicity (not in pregnancy)
- Best ACEI for CHF: ramipril, enalapril, and lisinopril
- Aliskiren: direct renin inhibitor (similar adverse effects)

ACEI-induced Angioedema

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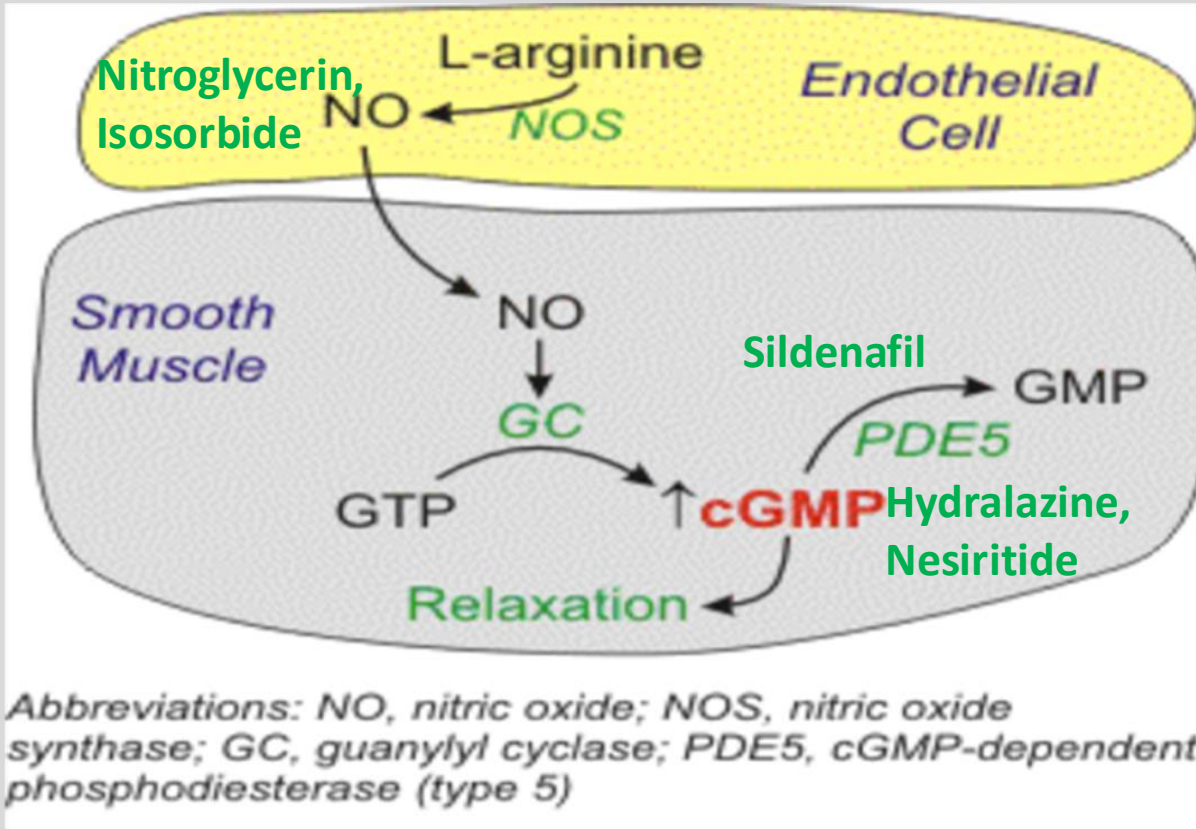
- Remove offending drug
- Treatment: epinephrine, glucocorticoids, antihistamines
- Icatibant: a synthetic bradykinin receptor antagonist for hereditary angioedema



Nitric oxide (NO) donors

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Nitric oxide (NO) donors

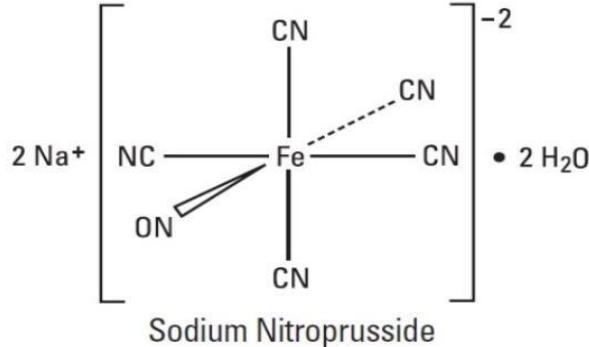
- **Act at the venule**
- **Decrease preload**
- **Decrease O₂ consumption**
- **Adverse effects: tolerance, headache
hypotension (avoid PDE-5 inhibitors)**

Nitroprusside

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- NO donor, but also dilates arterioles.
- Decrease preload and afterload
- Cyanide toxicity (hydroxycobalamin is the antidote)



Hydralazine



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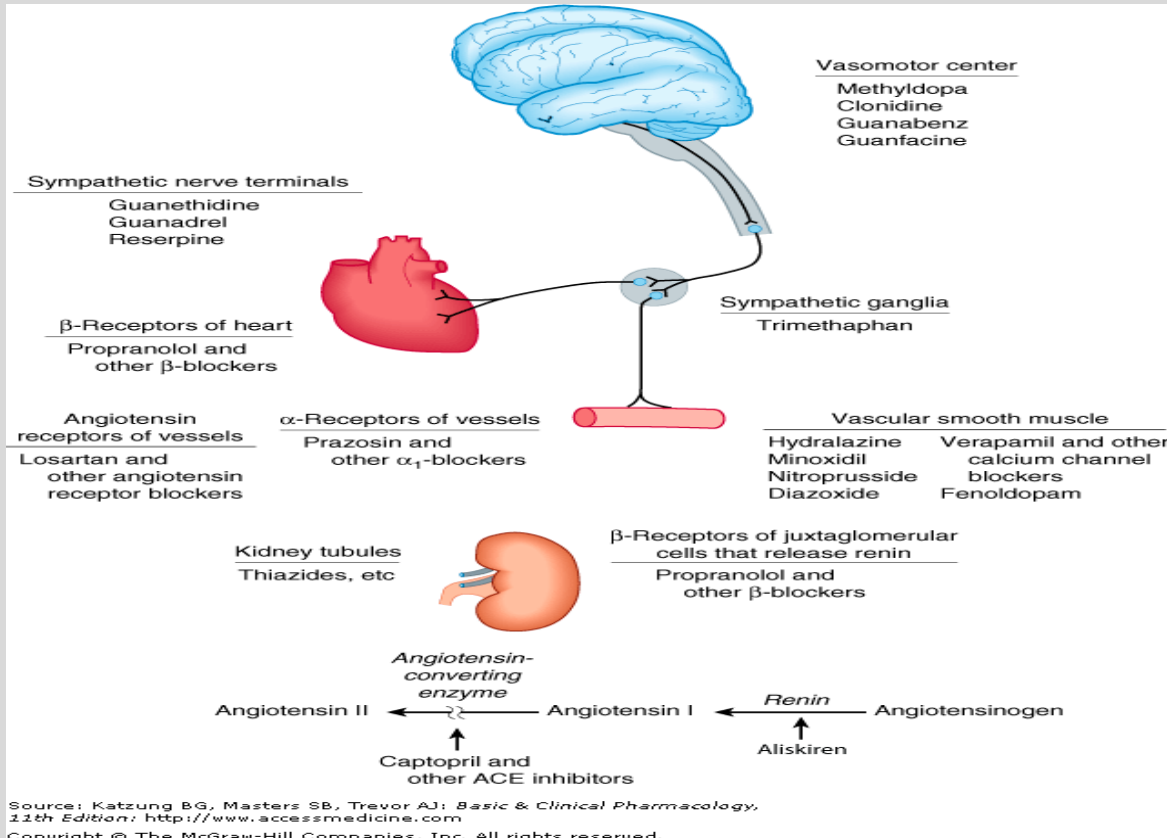
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- Arteriole (decrease afterload)
- Severe hypertension in pregnancy
- Acetylated (procainamide, isoniazid, sulfonamides)
- Adverse effects: headache, flushing, dizziness, hypotension, **tachycardia, fluid retention.**
- **Lupus-like syndrome** (slow acetylators)

Antihypertensives

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Beta-adrenergic antagonists

- Beta 1: decrease HR, CO, and renin from JXA
- Use: hypertension, MI, CHF
- Beta 1: (atenolol, metoprolol, bisoprolol)
- Beta 1, 2: (nadolol, propranolol, timolol).
Bronchoconstriction, hypoglycemia, PVD (Beta 2)
- Labetalol and Carvedilol (alpha and beta blocker, antioxidant potential)
- Pindolol and Acebutolol (ISA) Worsen!!!
- Esmolol (ultrashort acting)
- Nebivolol (nitric oxide effect)

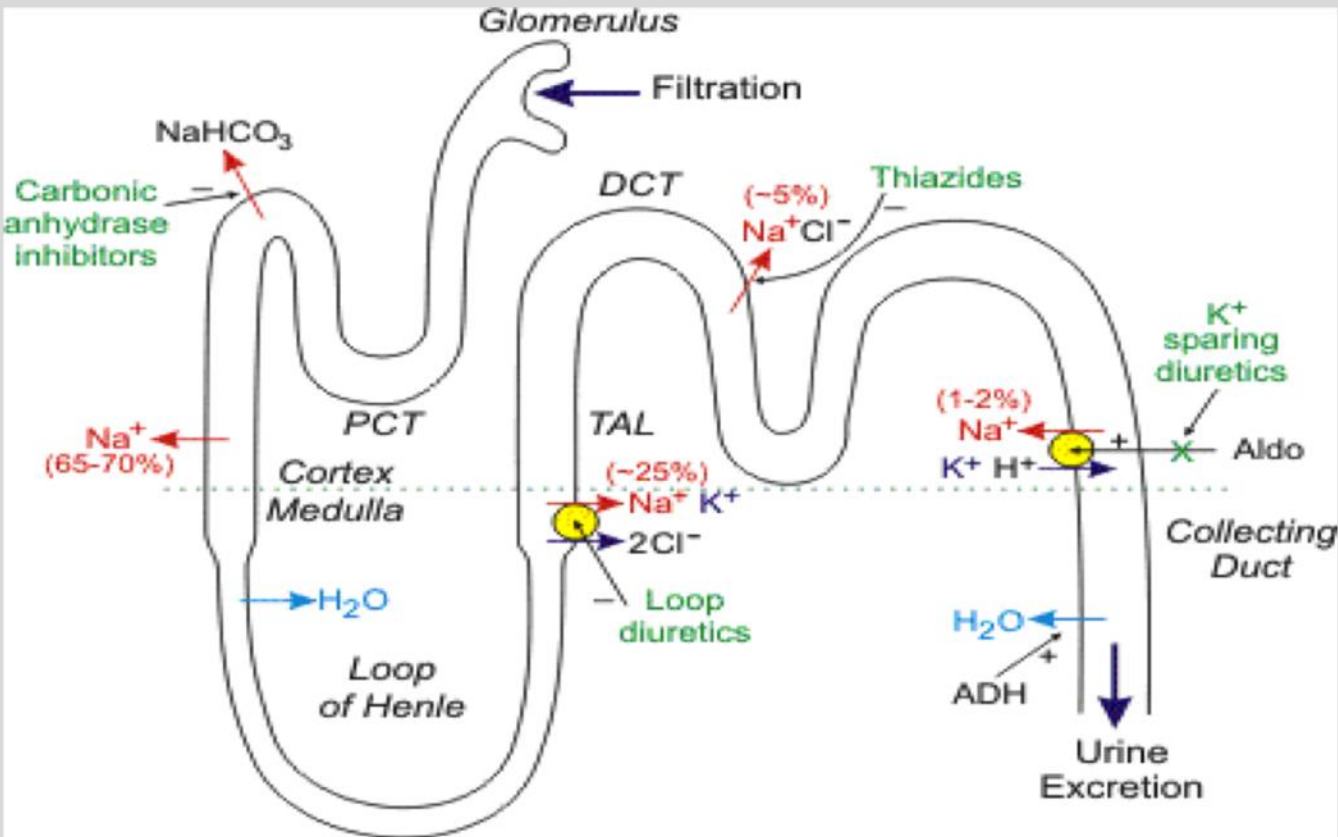
AV nodal suppressants

- **Calcium channel blockers (Class IV)
Verapamil and Diltiazem. (Not for CHF,
negative inotropic).**
- **Adenosine (Gi mechanism, for PSVT)**
- **Digoxin (vagus, parasympathetic)**
- **Beta adrenergic antagonists (Class II)**

Diuretics: Sites of Action

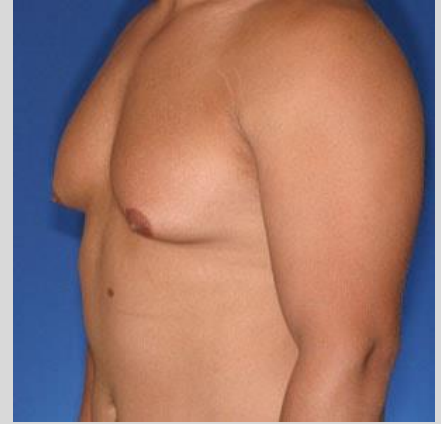
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Aldosterone Blockers (MRA)

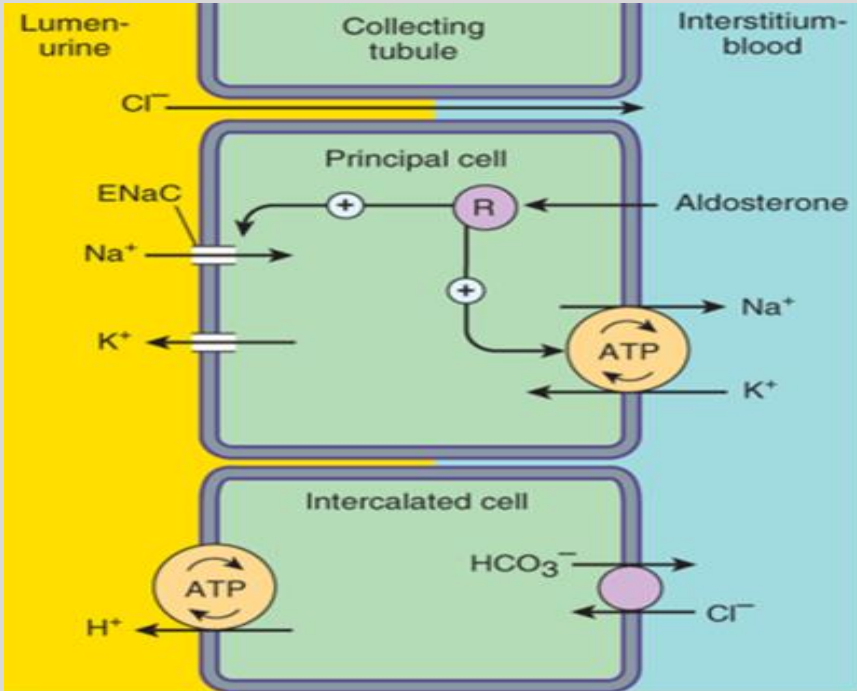
- Spironolactone, Eplerenone, Finerenone
- Antagonizes aldosterone receptors
- **Spironolactone (androgen blocker)**
- Use: high aldosterone/ CHF, acne, hirsutism
- Adverse effects: Hyperkalemia, acidosis
- Spironolactone: gynecomastia, impotence.
- Drug interactions: Digoxin, “renin” drugs, NSAIDs
- Finerenone: highly selective MRA. Slows progressive of CKD.



Aldosterone Blockers: Collecting Tubule

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Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology, 13th Ed.
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Loop Diuretics

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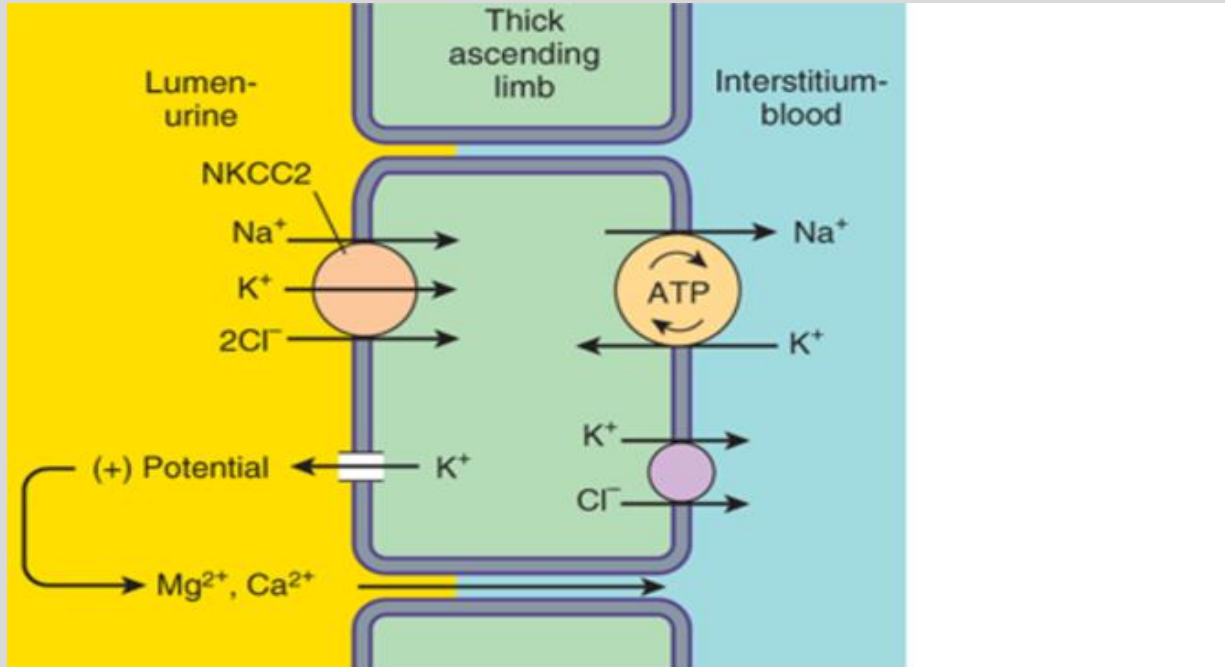
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- Furosemide, Bumetanide, Torsemide, **Ethacrynic acid (no sulfur)**
- Inhibit Na/K/2Cl- cotransport in thick ascending Loop of Henle (TAL).
- “High ceiling” diuretics
- Reduce preload

Loop Diuretics: TAL

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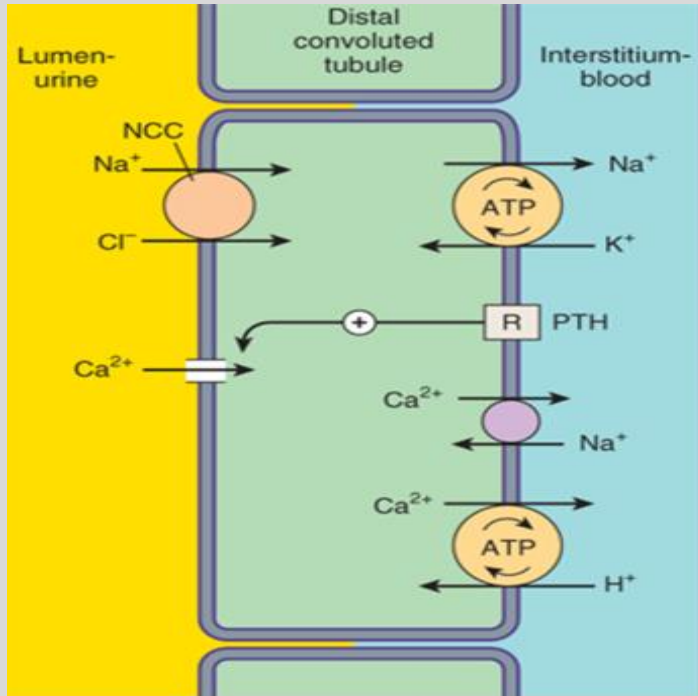
Loop Diuretics

- **Loss of Na, Cl, K, Ca, and Mg**
- **Increases PGE₂ (vasodilation)**
- **Use: CHF, pulmonary edema, hypercalcemia, hyperkalemia, hypertension, acute renal failure**
- **Adverse effects: hypokalemia, hypomagnesemia, hypocalcemia, hyperuricemia, metabolic alkalosis. Ototoxicity (especially with ethacrynic) hypersensitivity (except ethacrynic acid)**
- **Drug interactions: NSAIDs, digoxin, overlap ototoxicity (cisplatin, gentamicin, vancomycin)**

Thiazide Diuretics: DCT

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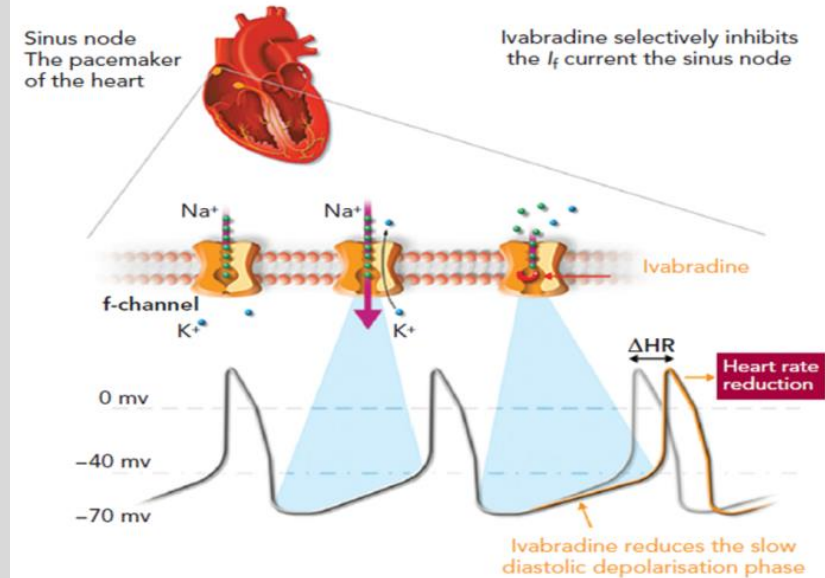
Thiazide Diuretics

- **Hydrochlorothiazide, Chlorthalidone, Indapamide**
- **Use: hypertension, CHF, kidney stones, and nephrogenic diabetes insipidus**
- **Adverse effects: Hypokalemia, alkalosis, hyperglycemia, hypercalcemia, hyperlipidemia, hyperuricemia, hypovolemia, hyponatremia, sulfa sensitivity.**
- **Drug Interactions: digoxin, NSAIDs, lithium**

Ivabradine

- Selective I_f inhibitor (“funny channel” blocker)
- Inhibits action in the SA node without affecting the duration of the action potential
- Use: CHF and stable angina, especially those who can't tolerate or be managed with beta antagonist
- Adverse effects: bradycardia, vision changes, not in pregnancy. Caution with CYP 3A4 inhibitors

Figure 1: Mechanism of Action of Ivabradine



Source: <http://www.shift-study.com/ivabradine/mode-of-action/> Reproduced with the permission of Servier © 2016.

Other Drugs

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- Angiotensin receptor/neprilysin inhibitor (ARNI): Sacubitril: (prodrug) inhibits neprilysin to enhance natriuretic peptides. Combined with Valsartan. Adverse effect: hypotension, hyperkalemia.
- Sodium-glucose cotransporter 2 inhibitor (SGLT2): dapagliflozin
- Conivaptan: blockade of V_{1a} receptor decreases PVR. V_2 blockade causes water secretion.
- Semaglutide: obesity related heart failure.

Subtype 2 sodium-glucose transport protein (SGLT2) inhibitors

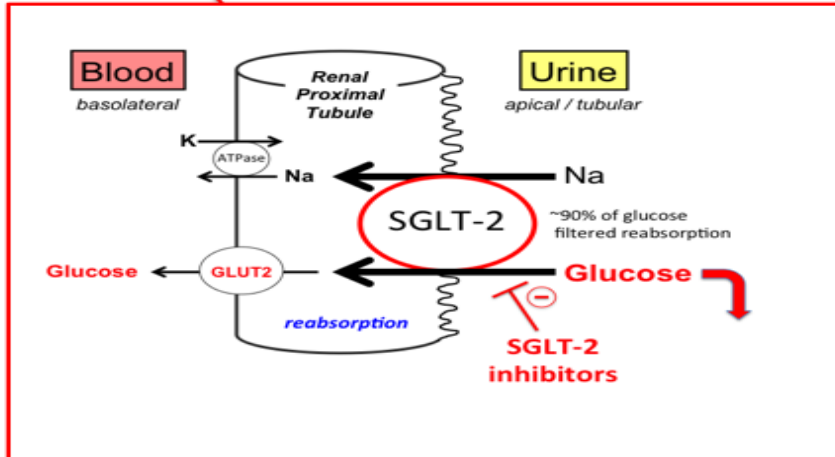
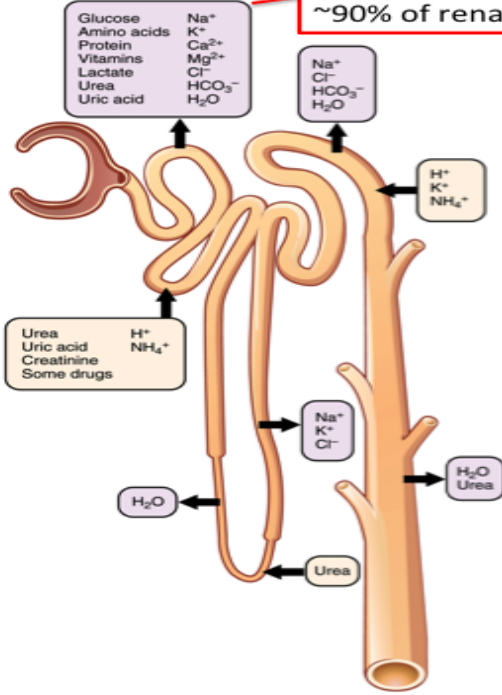
- Canagliflozin, **dapagliflozin, empagliflozin**
- Glucose excreted by kidney via proximal tubule
- CHF management: promotes diuresis.
- ~ 1% A1c reduction
- Adverse Effects: UTIs, monitor GFR (not for CrCl <30)

SGLT2 Inhibitors

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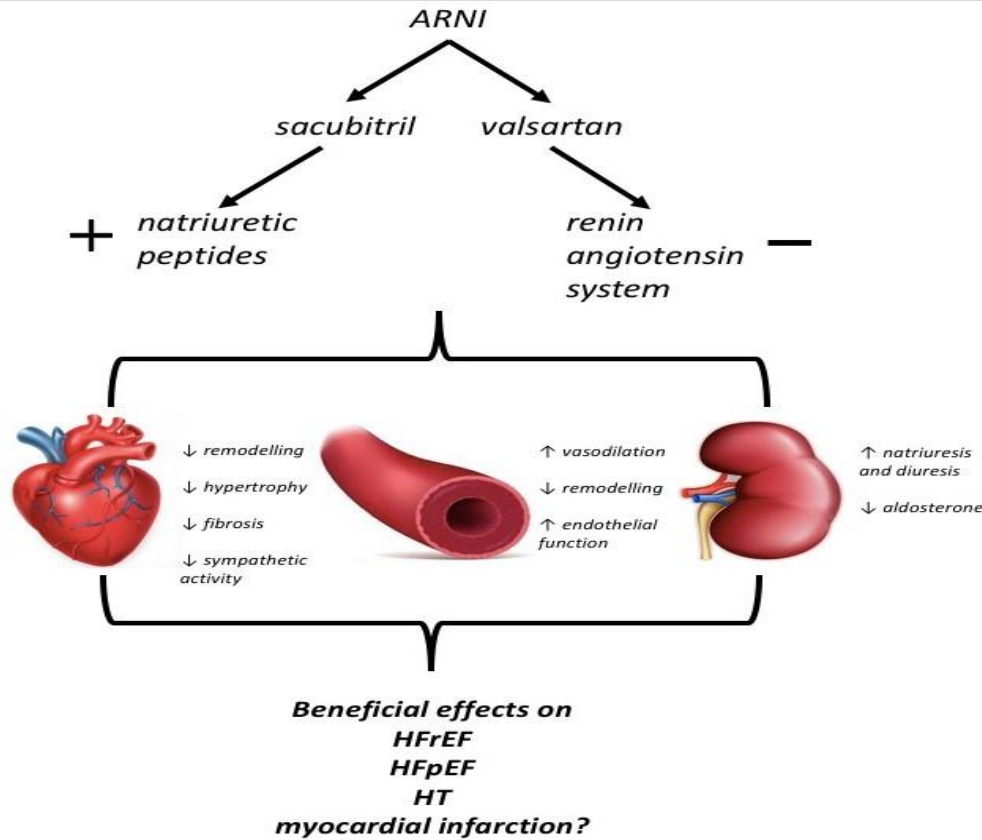
S1 segment proximal tubule:
~90% of renal glucose reabsorption



ARNI

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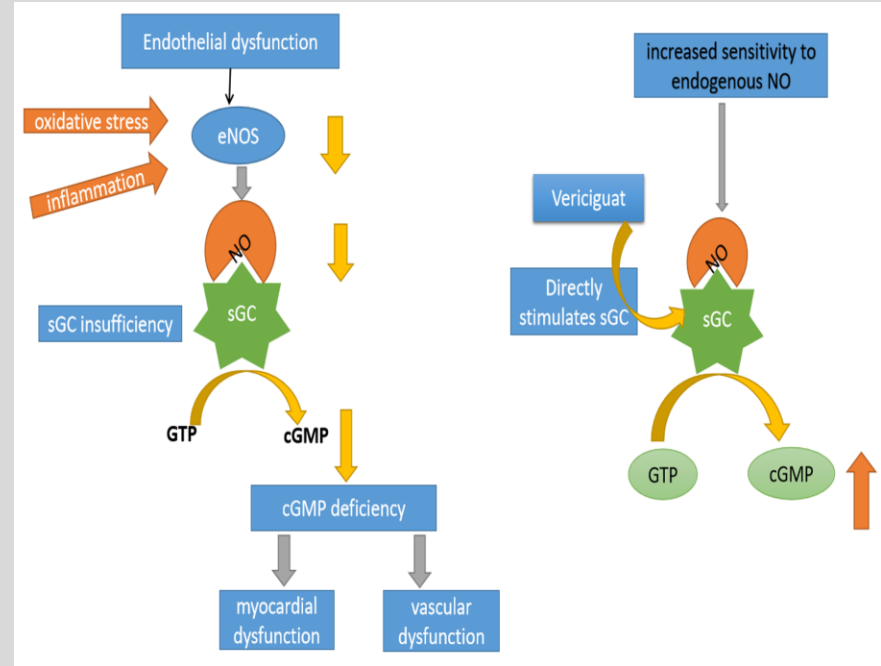
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Vericiguat

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- Stimulates soluble guanylate cyclase that activates the cGMP pathway independent of nitric oxide
- Adverse effects: Headache, dizziness, hypotension, tachycardia. Not to be taken with NO donors or PDE5-
- Does not induce tolerance.
- Reduced hospitalizations in patients with worsened chronic CHF (HFrEF)



Pharmacologic therapy of HFrEF

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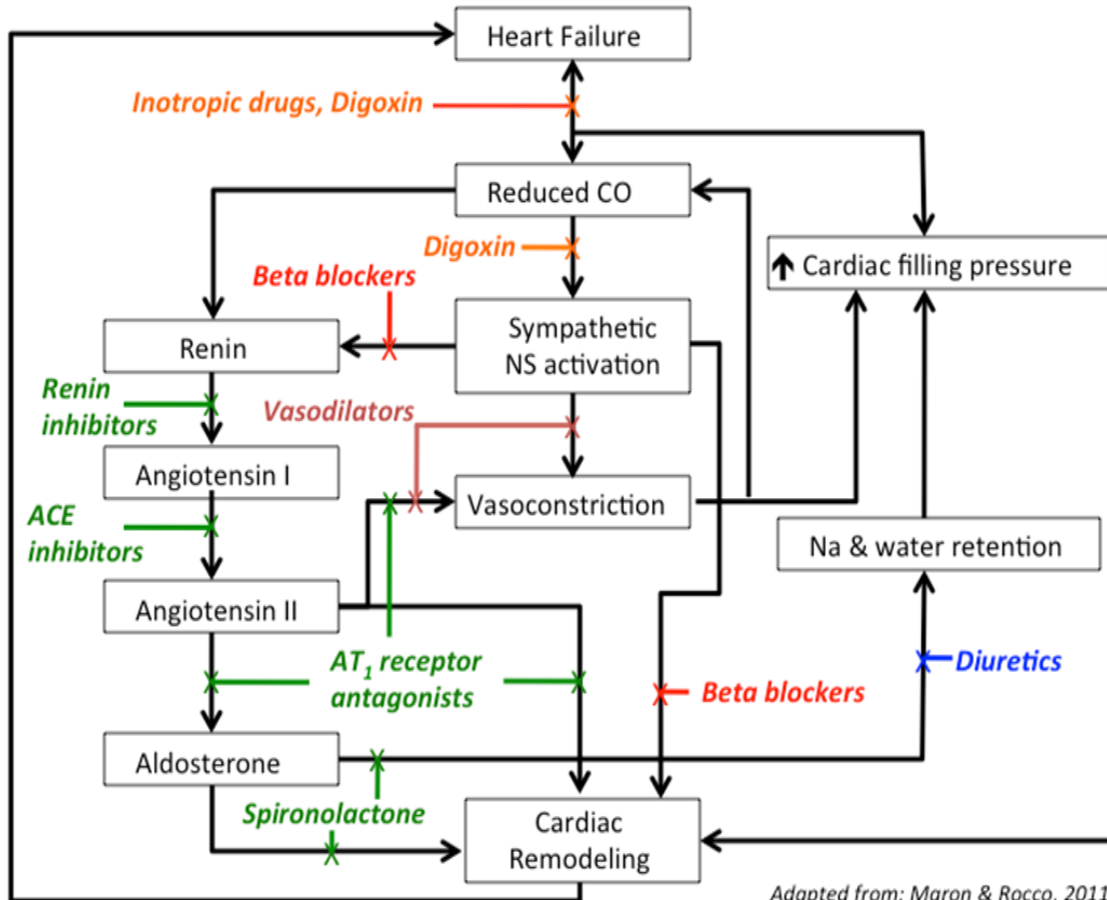
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- Improvement in symptoms: diuretic, beta blocker, ACEI (or ARB), ARNI, SGLT2-, hydralazine plus nitrate, digoxin, and mineralocorticoid receptor antagonist (MRA).
- Prolongation of patient survival: beta blocker, ACE inhibitors (or ARB), ARNI, dapagliflozin, hydralazine plus nitrate, and MRA.
- Limited evidence of survival benefit is available for diuretic therapy.

CHF Summary

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Summary Slide

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- **Correlate drugs for CHF management to its pathophysiology**

Core References:

- **Katzung & Trevor Basic and Clinical Pharmacology, 16e, 2024; Chapter 13: Drugs for Heart Failure; Chapter 12: Vasodilators; Chapter 15: Diuretics**
- **Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 33; Therapy of Heart Failure.**
- **Summary of CHF Drugs (Maria A. Pino, Ph.D)**
- **Scholar Bricks: CPG**

Review Q

A 66-year-old female was diagnosed with ventricular arrhythmia and started on medication. Several months later, she presented with excessive fatigue, weight gain, hair loss, and dry skin. Laboratory studies revealed a low T4 and a elevated TSH. Which drug induced this presentation?

- A. Disopyramide**
- B. Lidocaine**
- C. Propranolol**
- D. Amiodarone**
- E. Diltiazem**

Lecture Feedback Form:

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